Part A: File System Navigation

1. What is the **absolute path** to the file `essay3.docx` given the file system tree below?

```
~
|-- fall
|   |-- math53
|   |   |-- syllabus.pdf
|   |   |-- midterm.pdf
|   |   |-- final.txt
|   |-- stat20
|   |   |-- syllabus.docx
|   |   |-- labs
|   |       |-- lab1.qmd
|   |       |-- lab2.qmd
|   |       |-- lab3.qmd
|   |-- history7B
|       |-- syllabus.pdf
|       |-- essay1.txt
|       |-- essay2.txt
|       |-- essay3.docx
```

2. What is the **relative path** from the directory `history7B` to the file `essay1.txt`?

3. If you are inside the **stat20/labs** folder, what is the relative path to `math53/final.txt`?

4. If you are inside the **math53** folder, how can you print the content of `history7B/essay1.txt` using **only one line of code**?

5. If you are inside the **math53** folder, how can you create a new folder named `spring` in your **home directory ~** using **only one line of code**?

6. If you are inside the **stat20** folder, how can you copy file `lab1.qmd` into your newly created **spring** folder using **only one line of code**?

Part B: File System in R (use the same tree above)

7. Starting from your **home directory** `~`, write R code to **change the working directory** to `fall/math53`.

8. After changing the working directory, write the R command to display your current working directory.

9. Assume your **current working directory** is `stat20/labs`. Write R code to read the file `math53/final.txt` line by line into an object called `final`.

Part C: Working with Vectors and Matrices

10. Write an R command that produces the vector **(you cannot just list them directly)**:

```
[1]  2  4  6  8 10 12 14 16
```

11. Predict the output of this code:

```r
1:4 + 1:2
```

12. Predict the output of this code:

```r
c(4,5,6)^c(1,2)
```

13. Predict the output of this code:

```r
mean(c(T,T,FALSE,T,FALSE))
```

14. Using `z <- c("apple" = 2, "banana" = 4, "cherry" = 8)`, write the subsetting command that returns:

```
cherry banana  apple
     8      4      2
```

(note: there are multiple ways to do this)

15. Using `z <- c("apple" = 2, "banana" = 4, "cherry" = 8)`, write the subsetting command that returns:

```
banana banana
     4      4
```

(note: there are multiple ways to do this)

16. What is the data type of the vector `TGIF <- c("movies", "chicken wings", 100L, NA , NULL)`?

17. Write the R command that creates a dimension `3x2` matrix object **named** `mtx1`, using the vector `c(2,3,4,5,3,4)`.

18. For the `mtx1` matrix you created in question 17, write the R command that reassign the element `5` to `2` (note: there are multiple ways to do this).

19. Predict the result (write it as a matrix) of the R command:

```r
matrix(c(2,4,6,8),2,2)/2
```